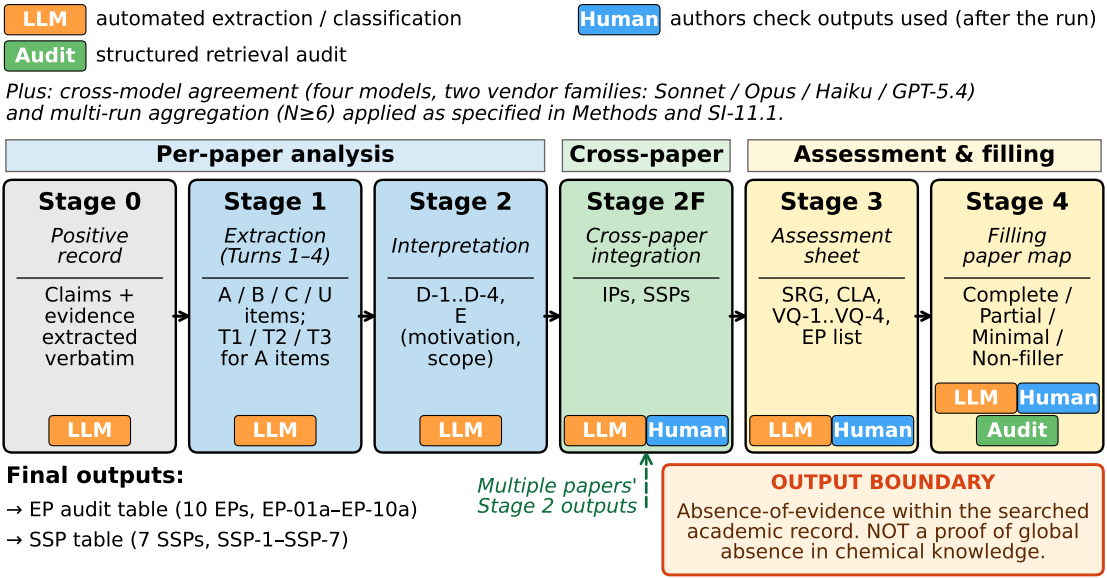

Governance dimensions (chip key):
LLM
automated extraction / classification
Human
authors check outputs used (after the run)
Audit
structured retrieval audit
Plus: cross-model agreement (four models, two vendor families: Sonnet / Opus / Haiku / GPT-5.4)
and multi-run aggregation (N≥6) applied as specified in Methods and SI-11.1.
Per-paper analysis
Cross-paper
Assessment & filling
Stage 0
Positive record
Claims + evidence extracted verbatim
LLM
Stage 1
Extraction (Turns 1–4)
A / B / C / U items; T1 / T2 / T3 for A items
LLM
Stage 2
Interpretation
D-1..D-4, E (motivation, scope)
LLM
Stage 2F
Cross-paper integration
IPs, SSPs
LLM
Human
Stage 3
Assessment sheet
SRG, CLA, VQ-1..VQ-4, EP list
LLM
Human
Stage 4
Filling paper map
Complete / Partial / Minimal / Non-filler
LLM
Human
Audit
Final outputs:
→ EP audit table (10 EPs, EP-01a–EP-10a)
→ SSP table (7 SSPs, SSP-1–SSP-7)
Multiple papers' Stage 2 outputs
OUTPUT BOUNDARY
Absence-of-evidence within the searched academic record. NOT a proof of global absence in chemical knowledge.
EP = Evidence Pack; SSP = Subgroup-Specific Proposition. A / B / C / U = absence categories;
T1 / T2 / T3 = inference tiers (A items only); VQ = Validation Quality; chips show governance.